

sEMG Prosthetic Hand - Iron Man Gauntlet

Introduction

Objective: Building an affordable, real-time prosthetic hand capable of executing 8+ gestures via user muscle intent (sEMG signals) through 4-channel MindRove armband & GUI.

We are modeling paradigms of technological accessibility through BCI projects and advancements in the supporting softwares. We can extend the framework of Iron Man Gauntlet to applications such as:

- Enhancing prosthetic accessibility and cooperative HCI
- Integrating smart device accessibility more widely
- Improving prosthetic control features
- Elevating biomimetic design and biomedical applications

Background Information

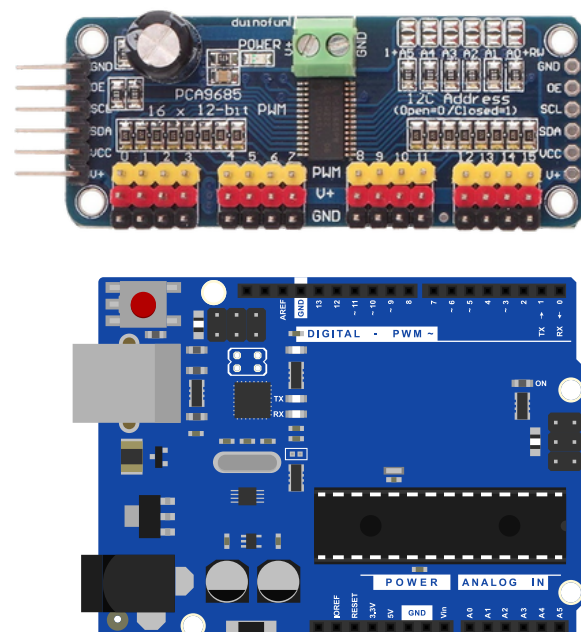
BCI: Brain-Computer Interface → Systems that convert brain signals into device commands.

sEMG: Surface electromyography → non-invasive muscle-neuron activity measurement tool.

Prosthetics: artificial limbs (cosmetic, body-powered, myoelectric, or hybrid) meant to assist amputees with quality of life.

Tools & Hardware

- Mindrove: 8-Channel EMG Armband
- Arduino Uno R3 Microcontroller
- 16 channel Servo Driver Module
- 5 Servos



Project Leads
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GUI DEMO

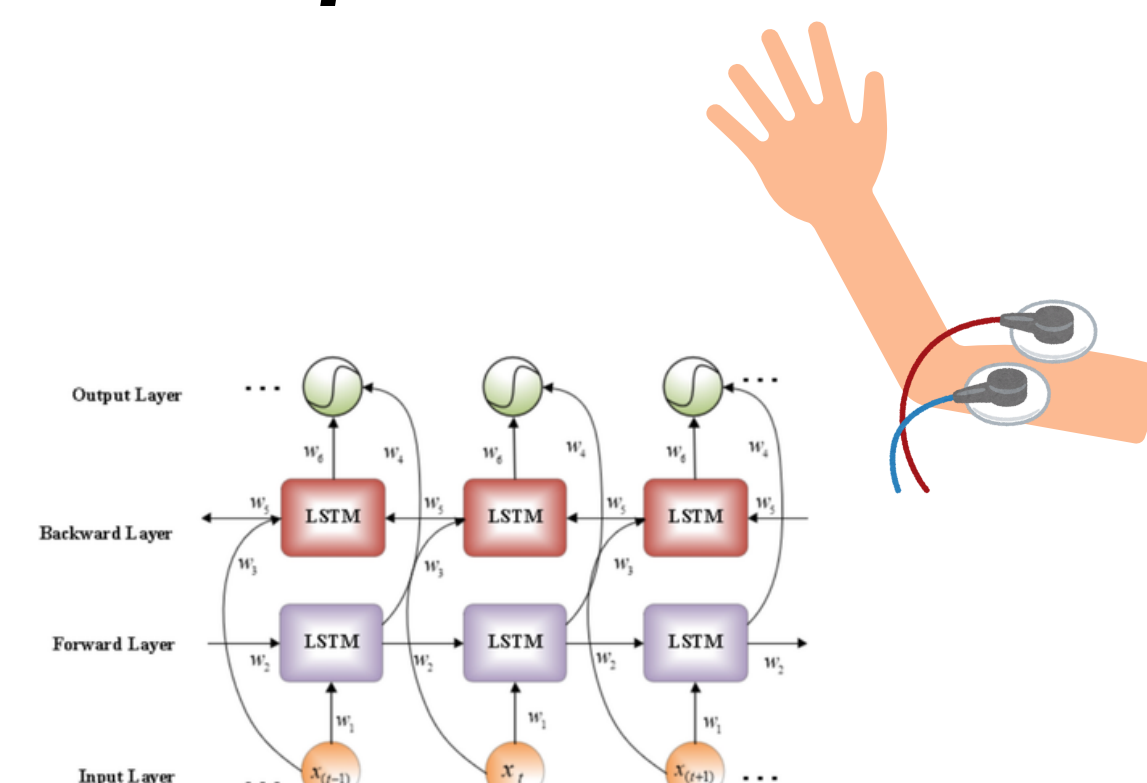


Project Outline



Final Prosthetic Arm

Arm receives commands and performs desired motion



sEMG signals recorded from electrode armband

Signals undergo filtering & preprocessing

Send commands to prosthesis wirelessly

Match user input to specified hand gesture commands

Train and implement deep learning model

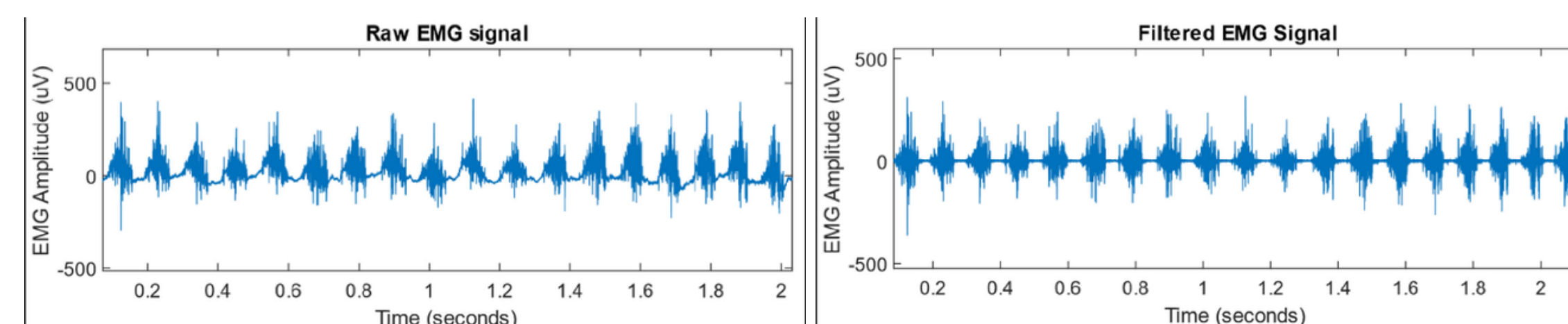
Our project follows this pipeline to transform biosignals captured from the sEMG armband to specified gesture prosthetic hand commands:

- The MindRove armband detects 4-channel sEMG signals at the forearm muscles triggered by user intent (hand contraction/extension).
- The GUI collects raw sEMG data which is filtered to isolate frequencies of interest and remove electrical noise.
- Signals are segmented into overlapping windows and normalized.
- Bi-LSTM automatically extracts features and classifies the gesture.
- GUI sends specific servo motion commands via Arduino over WiFi to the servos in the fingers for live control.
- The prosthetic executes the desired motion, moving fingers up and down and so forth.

Data Processing

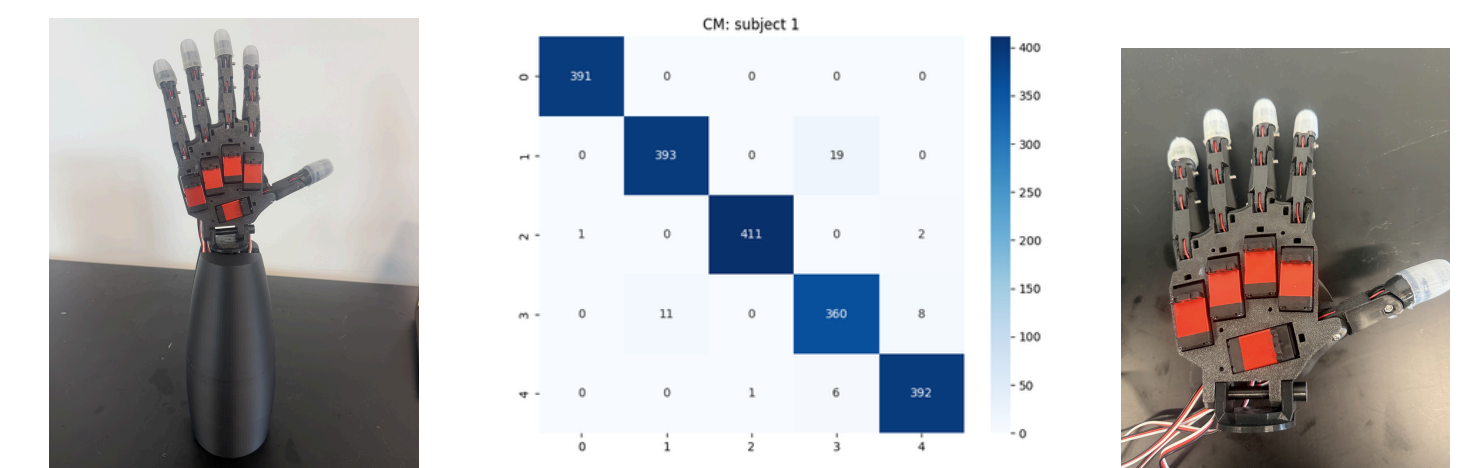
Using the MindRove SDK with BoardShim, sEMG signals undergo:

- IIR Notch Filter (60 Hz, Q = 30)
- Butterworth Bandpass Filter (10Hz - 200Hz, 4th Order)
- Overlapping Sliding Window & Normalization



Results and Analysis

- Functional Prosthetic Arm with fluid finger movement
 - Servo with tensioned cable+springs for control
 - Silicone fingertips/palm for biocompatibility
 - Modula arm component + wrist rotation
 - 50x cheaper than normal prosthetic arm
- Accessible GUI (collection, training, execution)
 - Developed sEMG processing & BiLSTM model
 - Automated feature extraction to identify distinct patterns and execute gestures via Arduino
 - 250ms prediction, 97% accuracy across users



Next Steps

- Enhanced wrist movement + added DOF
- Abduction/adduction for fingertips, wearability
- CNN-LSTMs, CNN-GRUs, or Transformers for complex spatial-temporal relationships in gesture recognition
- Proportional control algorithms for nuanced actuation
- Transfer learning protocols & model quantization for reduced calibration time & improved latency

References

Avilés-Mendoza, Karla, et al. "A 3D Printed, Bionic Hand Powered by EMG Signals and Controlled by an Online Neural Network." *Biomimetics*, vol. 8, no. 2, 14 June 2023, p. 255. PubMed Central, <https://doi.org/10.3390/biomimetics8020255>.

Karnam, Naveen Kumar, et al. "EMGHandNet: A Hybrid CNN and Bi-LSTM Architecture for Hand Activity Classification Using Surface EMG Signals." *Biomedical Signal Processing and Control*, vol. 42, no. 1, 2022, <https://doi.org/10.1016/j.bbe.2022.02.005>.